IBM FileNet P8 V5.5.8

Day 9

API Integration & Automation

Java CEI API

REST API

Event Automation

Java CEI | REST API | CMIS | Event-Driven Automation

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Learning Objectives – Day 9

API

Describe FileNet P8 V5.5.8 API landscape: Java CEI, .NET, REST, CMIS

API

Understand the CE Java API object model: Domain, ObjectStore, Document, Factory

REST

Use CMIS Browser Binding to upload documents and execute queries via HTTP

REST

Authenticate and call CE REST and PE REST endpoints with curl and JSON

Java

Write and compile Java CEI code to connect, create, update, and search documents

Java

Handle CEI exceptions and use PropertyFilter for efficient property retrieval

Events

Create Event Subscriptions in ACCE for OnCreation, OnCheckin, OnDelete

Events

Implement custom event handlers and Process Router workflow launchers

01

FileNet P8 API Landscape

Java CEI · .NET API · REST API · CMIS · Process Engine REST

FileNet P8 V5.5.8 – Supported API Types

Java CEI API

Full-featured Java API (EJB/MTOM). Deep object model access. Used by ACCE and ICN internally. JAR: Jace.jar

.NET / COM API

Windows .NET wrapper for Java CEI. C# / VB.NET integration. Same object model via FileNet.Api.dll.

CE REST / CMIS

Stateless HTTP/JSON. CMIS Browser Binding at /fncmis/browser. Ideal for web, mobile, and cross-platform apps.

PE REST API

HTTP API for workflow operations: launch, step, query work items. URL: /pe-rest/v1/. Used by ICN workflow UI.

Java CEI API – Object Model & Connection

Object Hierarchy

Connection

`Factory.Connection.getConnection(CE_URI)`

Domain

`Factory.Domain.fetchInstance(conn, null, null)`

ObjectStore

`Factory.ObjectStore.fetchInstance(domain, name, null)`

Document

`Factory.Document.createInstance(os, className)`

Folder

`Factory.Folder.fetchInstance(os, path, null)`

Properties

`doc.getProperties().putValue(key, value)`

MTOM Connection Code

```
// FileNet P8 V5.5.8 Connection
String CE_URI =
  "http://localhost:9080/wsi/
   FNCEWS40MTOM/";

Connection conn =
  Factory.Connection
    .getConnection(CE_URI);

Subject subj =
  UserContext.createSubject(
    conn, "ceadmin",
    "ceadmin", "FileNetP8");

UserContext.get()
  .pushSubject(subj);

Domain d = Factory.Domain
  .fetchInstance(conn,null,null);
```

Key Operations via Java CEI API

Document Creation

```
Factory.Document.createInstance(os,
"Document")
Set properties, attach ContentTransfer
checkin(MAJOR_VERSION), save(REFRESH)
File into folder via folder.file()
```

Metadata Update

```
Factory.Document.fetchInstance(os, id,
null)
getProperties().putValue("DocumentTitle",
value)
save(NO_REFRESH)
Supports String, Date, Integer, Boolean
types
```

Folder Operations

```
Factory.Folder.fetchInstance(os, path,
null)
Factory.Folder.createInstance(os, null)
set_FolderName() + set_Parent()
folder.get_ContainedDocuments() for
listing
```

Search Queries (CE SQL)

```
new SearchSQL("SELECT * FROM Document
WHERE...")
new SearchScope(os)
scope.fetchObjects(sql, maxItems, pf,
true)
Supports INFOLDER, LIKE, timestamp filters
```

02

REST API & CMIS Operations

CMIS Browser Binding · curl Commands · Upload · Search · PE Workflow

CE REST & CMIS API Endpoints (FileNet P8 V5.5.8)

Base URLs – Training Lab (localhost:9080)

CMIS Browser Binding `http://localhost:9080/fncmis/browser/`	Primary REST endpoint for all CRUD. Use ?cmisselector=repositories to get repo IDs.
CMIS AtomPub `http://localhost:9080/fncmis/resources/`	XML-based REST. Used by older CMIS clients and SharePoint connectors.
Process Engine REST `http://localhost:9080/pe-rest/v1/`	Workflow operations: launch, query, step through work items.
ACCE Admin Console `http://localhost:9080/acce`	Browser-based admin. Verify uploads, configure subscriptions.
Authentication `Authorization: Basic Y2VhZG1pbjpjZWFkbWlu`	Base64(ceadmin:ceadmin). Use -u ceadmin:ceadmin flag in curl.

REST API – Upload Document & Execute Search

Upload Document via CMIS Browser Binding (curl)

```
curl -u ceadmin:ceadmin -X POST -H "Accept: application/json" \
  -F "cmisaction=createDocument" \
  -F "propertyId[0]=cmis:name" -F "propertyValue[0]=MyDoc.txt" \
  -F "propertyId[1]=cmis:objectTypeId" -F "propertyValue[1]=cmis:document" \
  -F "content=@test_document.txt;type=text/plain" \
  "http://localhost:9080/fncmis/browser/{REPO_ID}/API_Labs/REST_Uploads"
```

Search for Uploaded Document (CMIS Query)

```
curl -u ceadmin:ceadmin -X POST \
  -H "Content-Type: application/x-www-form-urlencoded" \
  --data-urlencode "cmisaction=query" \
  --data-urlencode "statement=SELECT * FROM cmis:document WHERE cmis:name = 'MyDoc.txt'" \
  "http://localhost:9080/fncmis/browser/{REPO_ID}"
```

03

Event-Driven Automation

Event Subscriptions · Pre/Post Processing · Custom Event Handlers

Event-Driven Automation Architecture

Event

Predefined actions on FileNet objects.

OnCreation OnCheckin
OnDelete OnClassify
OnPromote OnDemote
OnChangeState

Always fired. Cannot be disabled.

Subscription

Links an Event on a Class or Folder to an Event Action.

Configured per Object Store in ACCE.

Can be scoped to specific folders.

Can be enabled or disabled.

triggers

Event Action

Code executed when subscription fires.

Types:
- Code Module (Java class)
- Process Router (launch workflow)
- Email Notification

Pre- or Post-processing.

invokes

Event Types & Pre/Post Processing

FileNet P8 V5.5.8 Event Types

OnCreation	New object created
OnCheckin	Version checked in
OnDelete	Object deleted
OnClassify	Auto-classification
OnPromote	Version promoted
OnChangeState	ACL modified

Pre vs Post Processing

PRE-PROCESSING (Synchronous)

Fires BEFORE commit to database

Can VETO the operation (throw exception)

Use for: validation, mandatory metadata

Risk: failure blocks the original operation

POST-PROCESSING (Asynchronous)

Fires AFTER commit (cannot veto)

Used for: notifications, audit, workflow launch

Failures do NOT affect the original operation

Recommended for most automation scenarios

Custom Event Handler – Code Module

```
public class DocCreationHandler
    implements IObjectChangeEventHandler {

  @Override
  public void onEvent(
      ObjectChangeEvent event,
      Id subscriptionId) {

    IndependentlyPersistableObject obj
        = event.get_Object();

    if (obj instanceof Document) {
      Document doc = (Document) obj;
      String title = doc.getProperties()
          .getStringValue("DocumentTitle");

      // Enforce mandatory metadata
      if (title == null || title.isEmpty())
        throw new EngineRuntimeException(...);

      System.out.println("Created: "+title);
    }
  }
}
```

Deployment as Code Module

1. **Compile**
   `javac -cp Jace.jar DocCreationHandler.java`
2. **Package**
   `jar cvf Handler.jar DocCreationHandler.class`
3. **Upload**
   ACCE > Events > Code Modules > New Code Module
4. **Action**
   ACCE > Event Actions > New > Type: Code Module
5. **Subscribe**
   ACCE > Subscriptions > New > Link to Event Action
6. **Test**
   Create a document of the subscribed class and verify

Day 9 Lab Exercises

LAB-09-01 · 30 min
Environment & Prerequisites Setup

- Verify CE V5.5.8 and CMIS endpoint
- Confirm curl available on command line
- Locate Jace.jar at C:\IBM\FileNet\ContentEngine\lib
- Create C:\Day9Labs working directory
- Create API_Labs folders in ACCE

LAB-09-02 · 45 min
Upload Document via REST API

- Get repository GUID via CMIS
- curl POST to createDocument in REST_Uploads
- Set cmis:name and cmis:description properties
- Verify upload in ACCE Browse
- Retrieve doc properties via REST GET
- Search with CMIS query statement

LAB-09-03 · 60 min
Java CEI API – CRUD Operations

- Compile FileNetConnect.java with Jace.jar
- Connect to CE V5.5.8 via MTOM transport
- Create document in Java_Uploads folder
- Update DocumentTitle via CEI API
- Execute CE SQL search and print results

LAB-09-04 · 45 min
Event Subscription & Workflow

- Create Process Router Event Action in ACCE
- Link to TrainingWorkflow + map F_Subject
- Create OnCreation Subscription scoped to API_Labs
- Test: create doc, verify workflow launches
- Query PE REST API for active work items

Day 9 Complete

API Integration & Automation

Java CEI API

Object model, connection, CRUD, search via Jace.jar

REST / CMIS API

CMIS Browser Binding, curl upload, CMIS query

Event Automation

Subscriptions, pre/post processing, code modules, workflow launch

Next: Day 10 – Deployment, Migration & Certification Preparation